Math 110 Hard Mock Midterm 2

Solutions

Question 1

(a)

Write $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Then $A(1,1)^T = (a+b,\ c+d)^T$, so

$$A \in U \iff a+b=0 \text{ and } c+d=0.$$

U is the null space of $A \mapsto A(1,1)^T$, a map onto $\mathbb{R}^2$ (it sends $E_{11} \mapsto e_1$ and $E_{21} \mapsto e_2$). By rank–nullity,

$$\dim U = 4-2 = 2.$$

W has basis $E_{11},\ E_{22},\ E_{12}+E_{21}$, so $\dim W = 3$.

For $U \cap W$: symmetric means $b=c$; then $a+b=0$ and $b+d=0$. So

$$U \cap W = \left\{ \begin{pmatrix} a & -a \\ -a & a \end{pmatrix} \right\}, \qquad \text{basis } \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

(b)

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W) = 2+3-1 = 4 = \dim V,$$

so $U+W=V$. The sum is not direct, because $U \cap W \neq \{0\}$.

Trap. A sum can be all of V without being direct. Here $2+3>4$ already forces a nonzero intersection.

(c)

Take $W' = \operatorname{span}(E_{11}, E_{22})$, the diagonal matrices. If a diagonal matrix $\operatorname{diag}(a,d)$ is in U, then $a+0=0$ and $0+d=0$, so it is 0. Hence $U \cap W' = \{0\}$. Since $\dim U + \dim W' = 2+2=4$, we get $V = U \oplus W'$.

Question 2

(a)

If $Sv=0$, then $Tv = R(Sv) = R(0) = 0$. So $\operatorname{null} S \subseteq \operatorname{null} T$.

(b)

Well defined. If $Sv = Sv' = y$, then $S(v - v') = 0$, so $v - v' \in \operatorname{null} S \subseteq \operatorname{null} T$. Hence $Tv = Tv'$: the value of $R_0(y)$ does not depend on the choice of v.

Linear. If $Sv = y$ and $Sv' = y'$, then $S(\lambda v + v') = \lambda y + y'$, so

$$R_0(\lambda y + y') = T(\lambda v + v') = \lambda Tv + Tv' = \lambda R_0(y) + R_0(y').$$

(c)

Take a basis $y_1, \ldots, y_r$ of $\operatorname{range} S$ and extend it by $x_1, \ldots, x_s$ to a basis of X. Define $R \in \mathcal{L}(X, W)$ on this basis by

$$R(y_i) = R_0(y_i), \qquad R(x_j) = 0$$

(LADR 3.4). R and R_0 are linear and agree on a basis of $\operatorname{range} S$, so they agree on all of $\operatorname{range} S$. Therefore $RSv = R_0(Sv) = Tv$ for every v, i.e. $T = RS$.

Trap. R_0 alone is not an answer. It is defined only on $\operatorname{range} S$, and the question asks for a map on all of X.

(d)

For $T(p) = p'(1) - p'(0)$, take

$$R(q) = \int_0^1 q(t)\, dt.$$

Then $R(Sp) = \int_0^1 p''(t)\, dt = p'(1) - p'(0) = T(p)$.

For $T(p) = p(0)$: the constant 1 is in $\operatorname{null} S$ (since $1'' = 0$), but $T(1) = 1 \neq 0$. So $\operatorname{null} S \not\subseteq \operatorname{null} T$, and by (a) no R exists.

Question 3

(a)

Images of the basis B:

$$T(1) = (1, 0, 1), \qquad T(x - 1) = (0, 1, 1), \qquad T\big((x-1)^2\big) = (0, 0, 1).$$

Coordinates in C: from $(\alpha, \beta, \gamma) = a(1,0,0) + b(1,1,0) + c(1,1,1) = (a + b + c,\ b + c,\ c)$,

$$c = \gamma, \qquad b = \beta - \gamma, \qquad a = \alpha - \beta.$$

Applying this to each image gives the columns:

$$\mathcal{M}(T) = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix}.$$

(b)

Both spaces have dimension 3, so it is enough to show T is injective. If $T(p) = 0$, then $p(1) = p'(1) = 0$, so $(x - 1)^2$ divides p: $p = c(x - 1)^2$. Then $p(2) = c = 0$, so $p = 0$.

(c)

Write $p = a + b(x-1) + c(x-1)^2$. Then

$$p(1) = a = 1, \qquad p'(1) = b = 1, \qquad p(2) = a + b + c = 1 \;\Rightarrow\; c = -1.$$

So

$$p = 1 + (x-1) - (x-1)^2 = -x^2 + 3x - 1.$$

Check: $p(1) = 1$; $p'(x) = -2x + 3$, so $p'(1) = 1$; $p(2) = -4 + 6 - 1 = 1$.

Question 4

(a)

$S = \pi \circ T$, where $\pi\colon W \to W/U$ is the quotient map. A composition of linear maps is linear. $S(v)$ is the zero of W/U exactly when $Tv + U = 0 + U$, i.e. $Tv \in U$, i.e. $v \in T^{-1}(U)$. So $\operatorname{null} S = T^{-1}(U)$.

(b)

$\operatorname{range} S = \{Tv + U\} = \pi(\operatorname{range} T)$. Since π kills U, also $\pi(\operatorname{range} T + U) = \pi(\operatorname{range} T)$. So $\operatorname{range} S$ is the range of π restricted to $X = \operatorname{range} T + U$. The null space of that restriction is $X \cap U = U$. Rank–nullity on X:

$$\dim \operatorname{range} S = \dim(\operatorname{range} T + U) - \dim U.$$

(c)

Rank–nullity for S, with (a) and (b):

$$\dim T^{-1}(U) = \dim V - \dim(\operatorname{range} T + U) + \dim U.$$

Substitute $\dim(\operatorname{range} T + U) = \dim \operatorname{range} T + \dim U - \dim(U \cap \operatorname{range} T)$ and $\dim V - \dim \operatorname{range} T = \dim \operatorname{null} T$:

$$\dim T^{-1}(U) = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T).$$

(d)

$\operatorname{null} D = \operatorname{span}(1)$, so $\dim \operatorname{null} D = 1$. $\operatorname{range} D = \mathcal{P}_3(\mathbb{R})$: derivatives of polynomials in $\mathcal{P}_4$ have degree at most 3, and every such polynomial is one. So $U \cap \operatorname{range} D = \operatorname{span}(x^3)$, of dimension 1. By (c),

$$\dim D^{-1}(U) = 1 + 1 = 2.$$

Basis: $p' \in \operatorname{span}(x^3, x^4)$ and $\deg p' \le 3$ force $p' = cx^3$, so $p = a + \frac{c}{4}x^4$. A basis is $1,\ x^4$.

Trap. $U \cap \operatorname{range} D$ is not all of U here, unlike the practice midterm, because the codomain is $\mathcal{P}_4$ and x^4 is not a derivative.